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Market Briefs Pro
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Happy Saturday, Pro Briefers!

This week we've identified an Innovation Shift. But before we tell you what it is, here's a little background information you should understand.

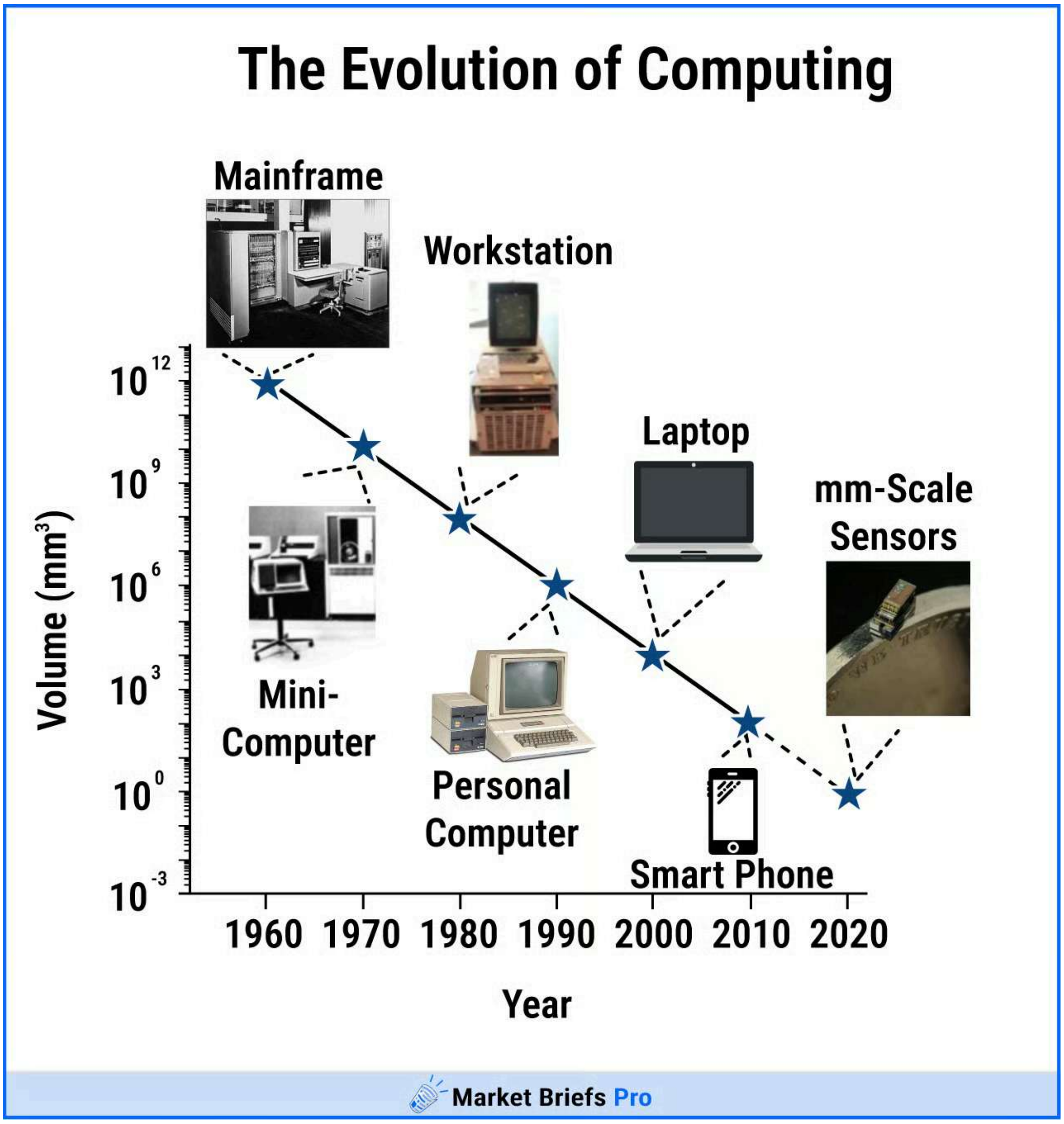
In the 90s, the average computer chip was around 50 millimeters in size, or about the size of a golf ball.

- It stored around 45 megabytes.

Computer chips have gotten smaller since then. In 2025, most computer chips are only around 25 millimeters in size, but they can hold up to a terabyte of storage.

1,048,576 megabytes = 1 terabyte.

But computer chips are getting even smaller.



Data via SNS Institute

How? *Molecular computing.*

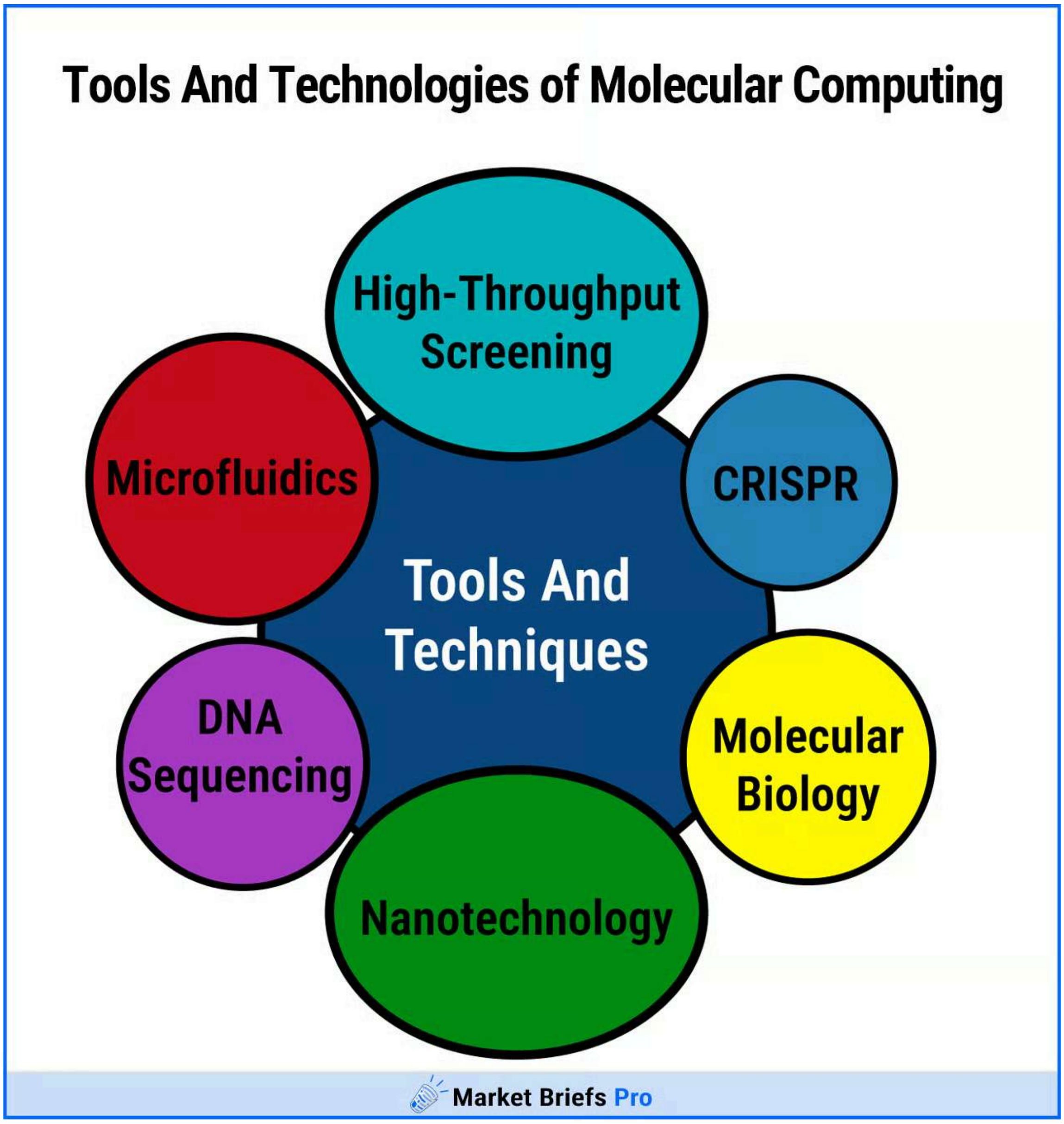
Molecular computing uses tiny computers that are 428 times smaller than the tip of a sewing pin.

But these tiny chips aren't for streaming videos - they are used to store data on our DNA.

This isn't just science, it's an opportunity some investors are looking at.

And, it's where we found a shift: New technology is opening up ways to invest in tiny tech, biology, and special chemicals.

- According to Straits Research, the market is expected to grow by 17% each year, from under $1 billion in 2025, to almost $4 billion by 2031.



Data via Biotech Research Asia

Last year, scientists at Harvard, MIT, and Caltech wrote more about molecular computers than then ever before.

To learn more about this, we got in contact with a bio-AI firm called Salt.ai.

We sat down for an exclusive interview with the CEO of Salt.ai, Aber Whitcomb, and the Head of Life Sciences at Salt.ai, Nate Beyor, to break this down.

"We've just seen an explosion in the rates of being able to do sequencing and DNA. And, that has really unlocked a lot of different opportunities." - Nate Beyor.

Why? It's cheaper than ever before to develop a molecular computer - costs over the last decade have fallen by 10,000%.

- Molecular computers use 10,000% less energy because they're super tiny.

In short - this micro innovation is shifting spending in a big way.

Let’s break down how investors may be able to take advantage of this opportunity.

We want your feedback: Tell us what you think about this edition of Market Briefs Pro by clicking here.

NANOTECH

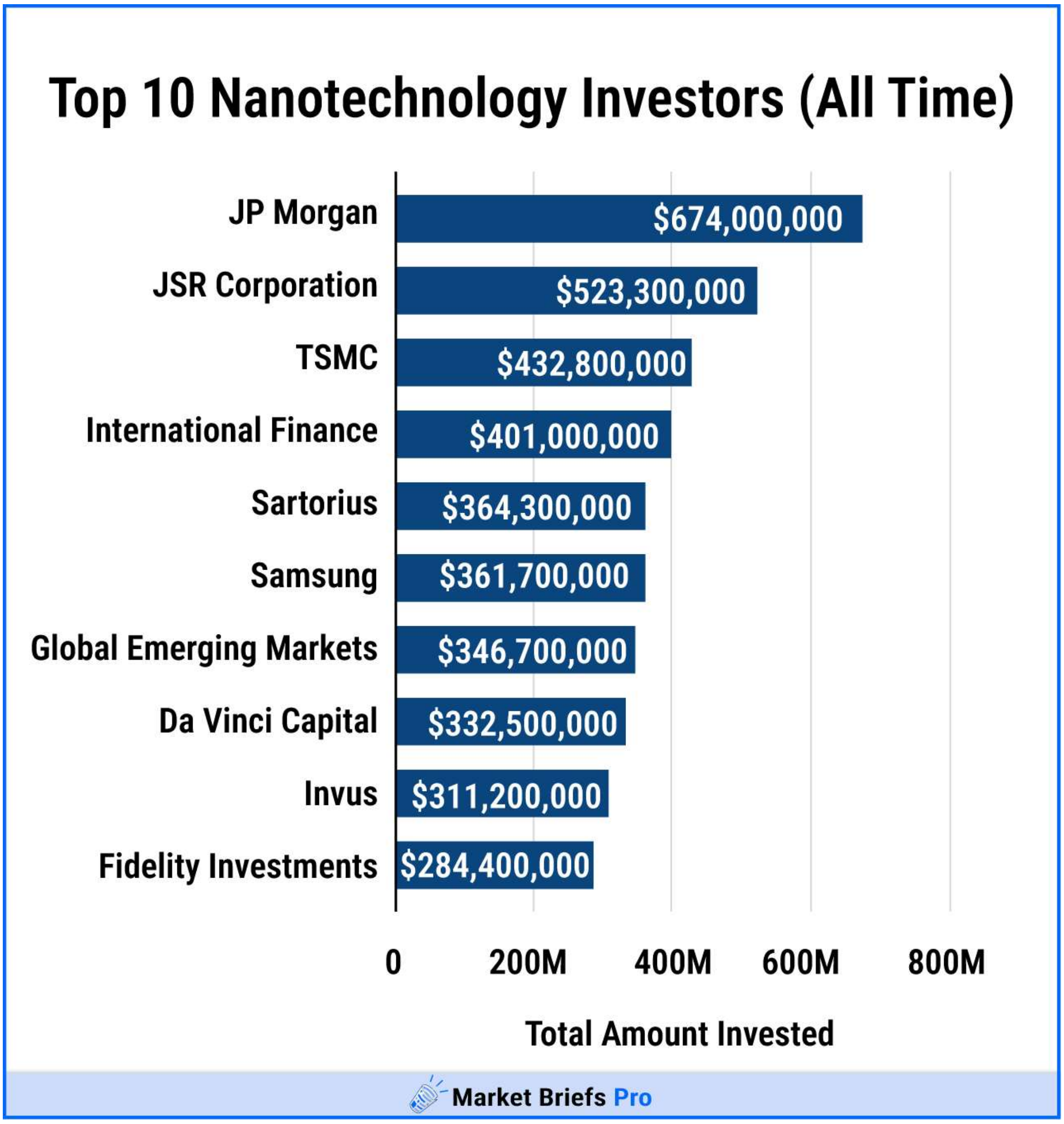
Creating Tiny Robots

According to the World Economic Forum, around 29% of the world’s labor is now generated by machines.

The point? Machines do a lot of the work that humans can do.

And this trend is entering healthcare.

Businesses are now using machines to do things medicine used to do, like fighting viruses and germs.



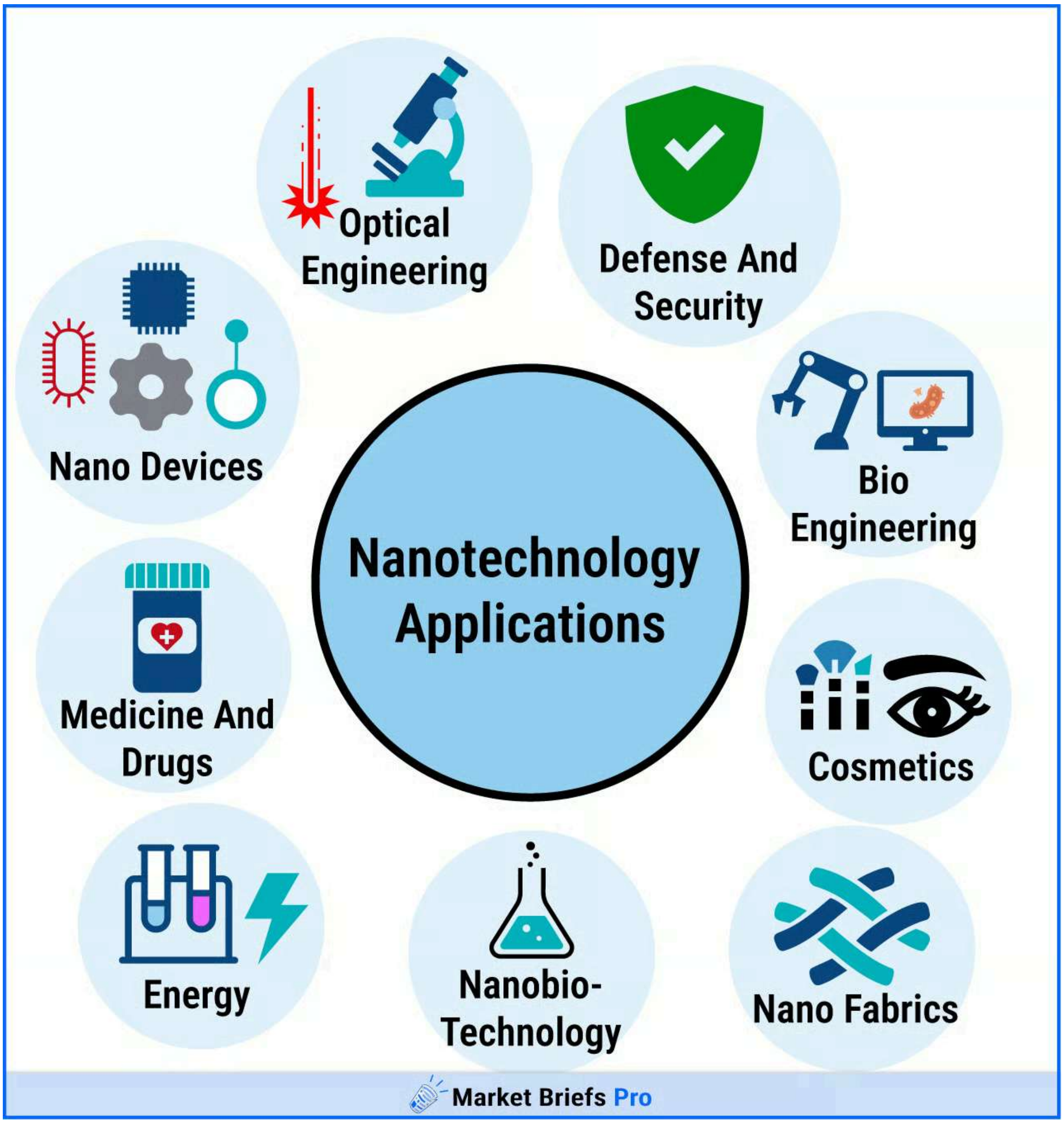
Data via StartUs Insights

For instance: Imagine building with LEGO blocks so tiny that you can't see them without a powerful microscope.

Data via New Atlas

Scientists use these tiny “blocks” to build tools that can go inside our bodies.

- For example, these tiny robots can swim through our blood vessels to deliver medicine directly to a tumor.



Data via Techovedas

"There's been massive innovation on the software side but, it wouldn't be possible without the innovation on the hardware side." - Aber Whitcomb.

So what companies are involved in this shift?

Applied Materials (AMAT) is supplying the building blocks scientists and doctors need to create these tiny robots.

One of the critical components of a molecular computer within these robots is semiconductor wafers.

Semiconductor wafer? These wafers are thin discs that computers are built on top of.

Last year, the creation of these wafers accounted for 74% of Applied Materials revenue.

- It reported $27 billion in revenue in 2024, a record high, and up 2% from the previous year.

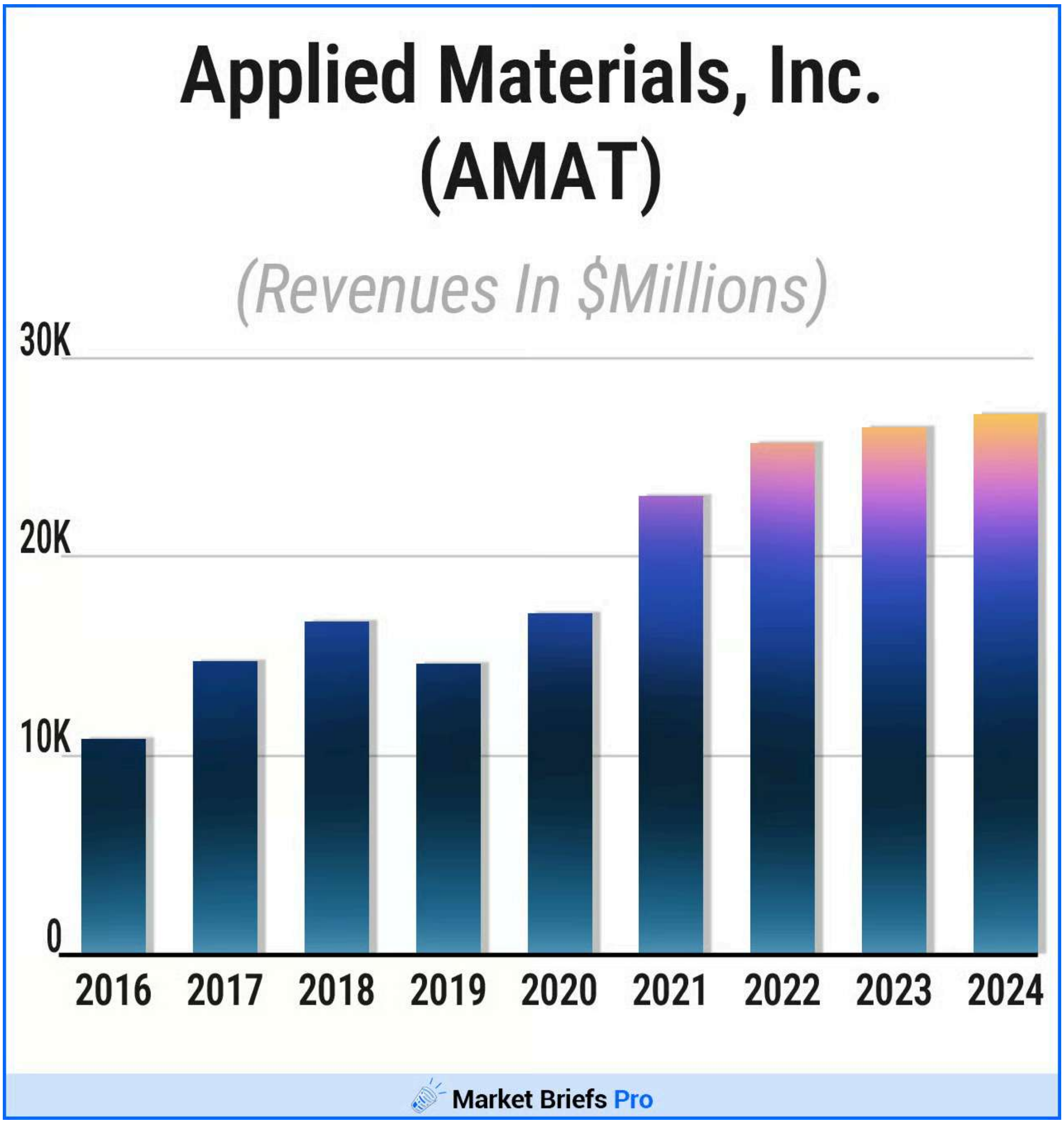
Data via Yahoo! Finance

Globally, it's one of the largest wafer semiconductor companies by market cap, just behind TSMC.

It installs and maintains wafers inside microscopic semiconductors.

27% of its operating revenue comes from equipment maintenance, which includes the wafers.

And as research and development grows for molecular computing, Applied Materials may also see its revenue rise alongside demand.



Data via company reports

Once the computers are built, they need to be installed in one of the microscopic robots.

Veeco Instruments (VECO) installs tiny computers on microscopic robots.

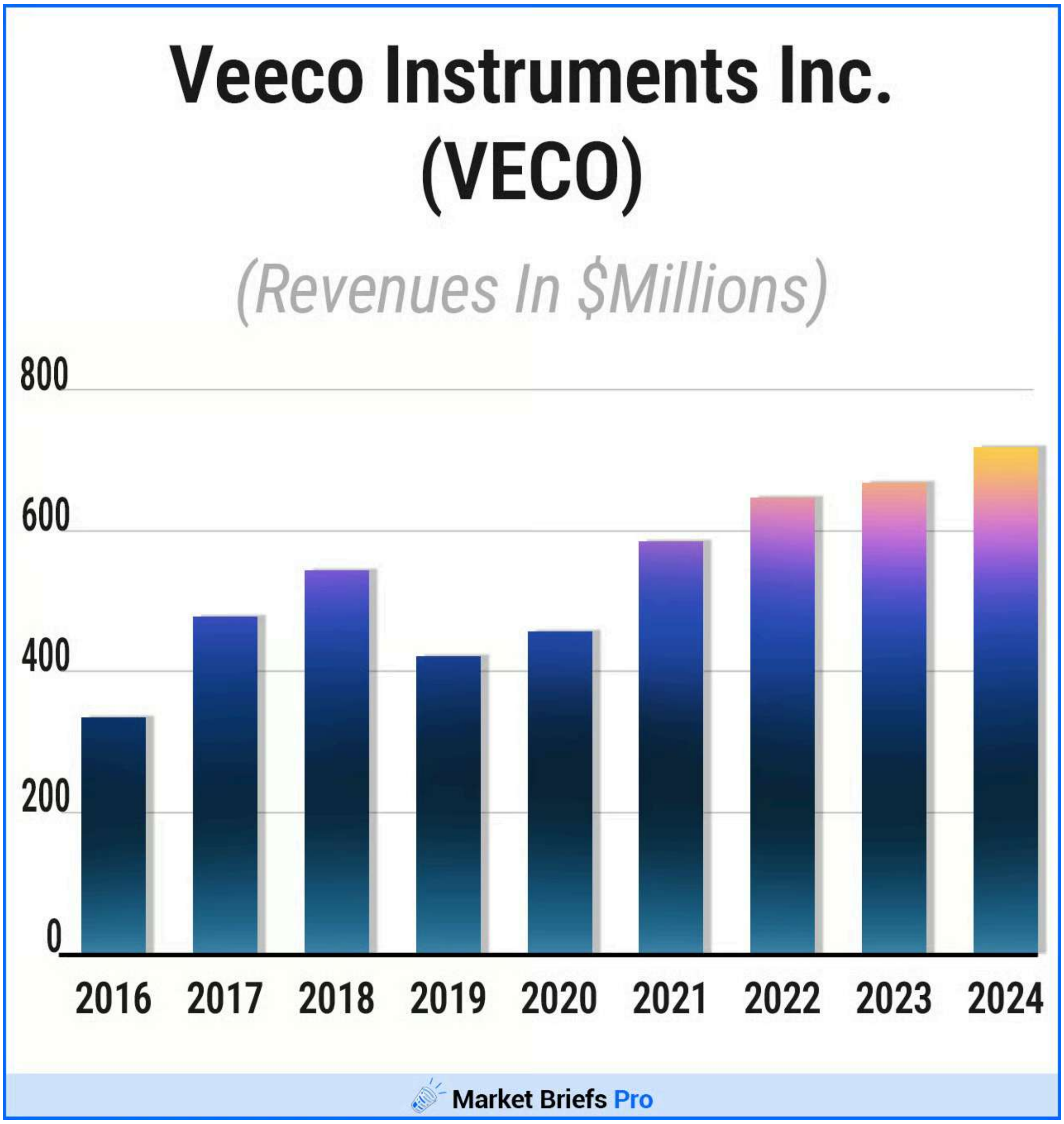
What does that mean? It works with scientists to create custom nanotech for labs to run experiments.

Data via company reports

And demand for nanotech has led to an increase in revenue for Veeco.

In 2024, the company reported $717 million in revenue, an increase from $666 million in 2023.

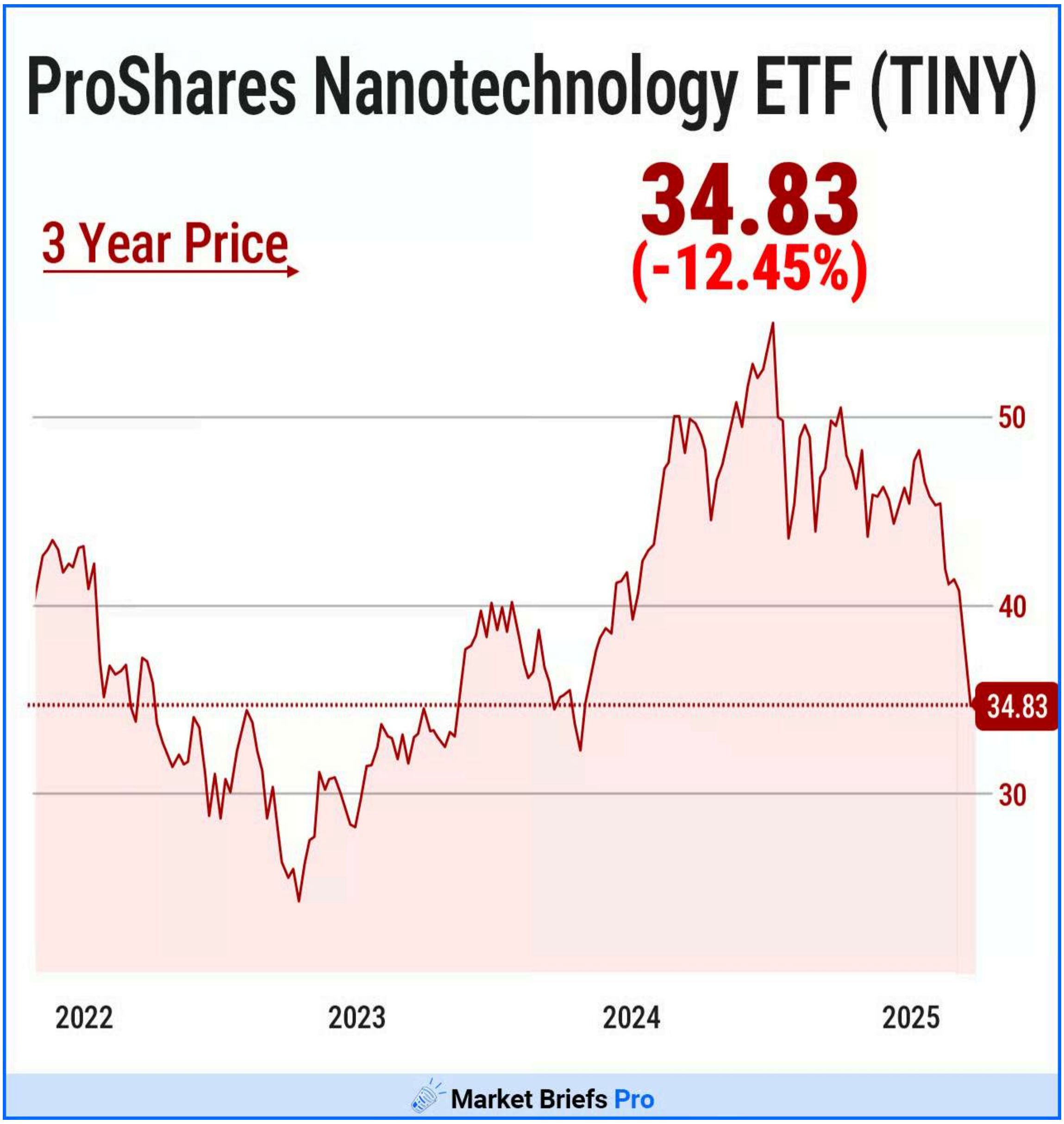
No other company is producing nanotechnology at the rate or scale that Veeco is, enabling them to take advantage of this Shift.



Data via Yahoo! Finance

- Veeco added 1,440 nanotech patents from 2009 to 2023, among the most of any nanotech company.

Investors looking for broad market exposure to the nanotechnology sector of the quantum computing industry may also find opportunities in the ProShares Nanotechnology ETF.



Data via Yahoo! Finance

BIOLOGY

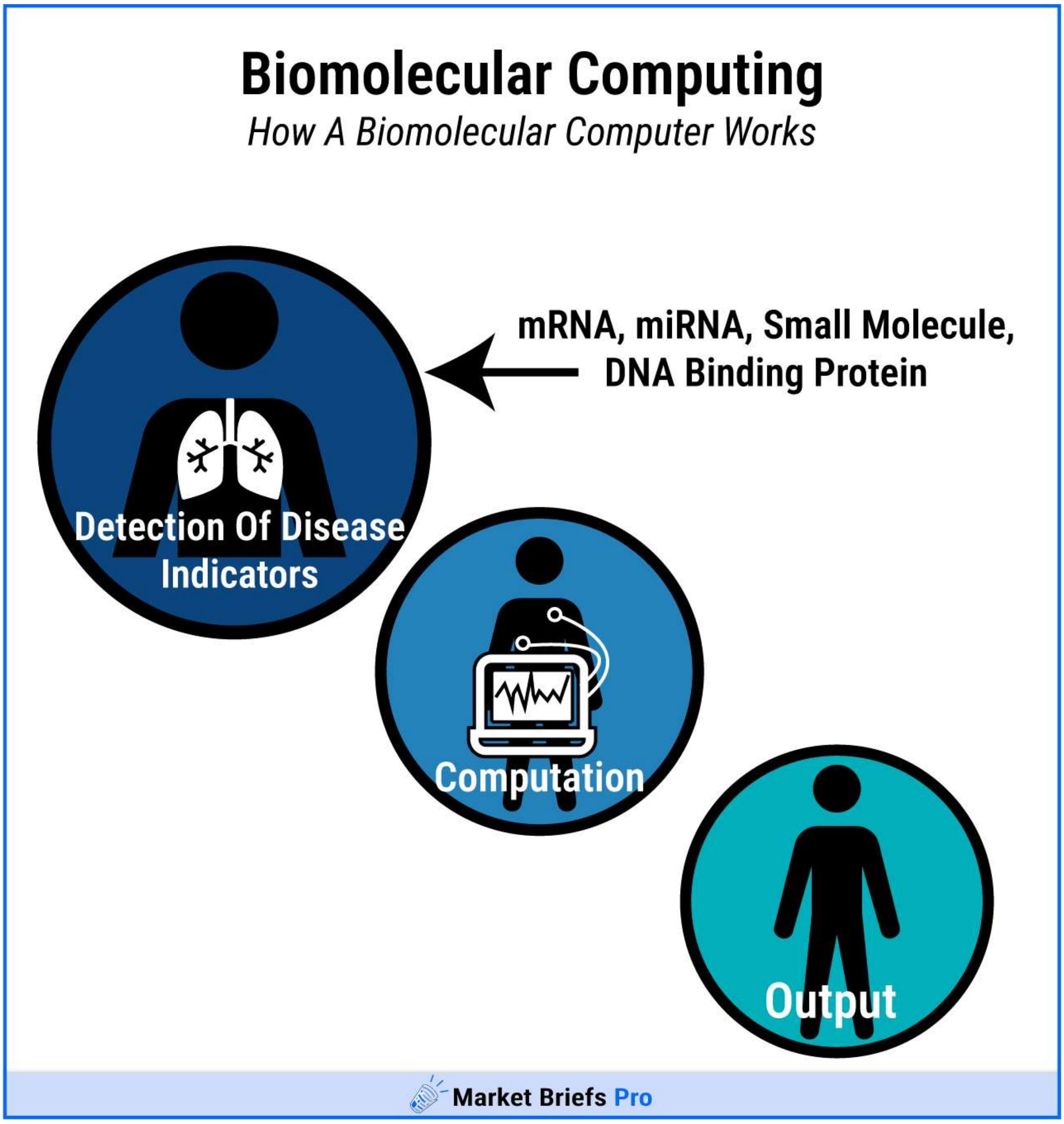
Programing DNA

Nanotech is hardware like computer chips and robots.

But you can't have a computer without software - that's where molecular biology comes in.

Biomolecular companies are working on molecular coding that can rewrite our DNA.

Think of DNA like nature's hard drive.



Data via Collidu

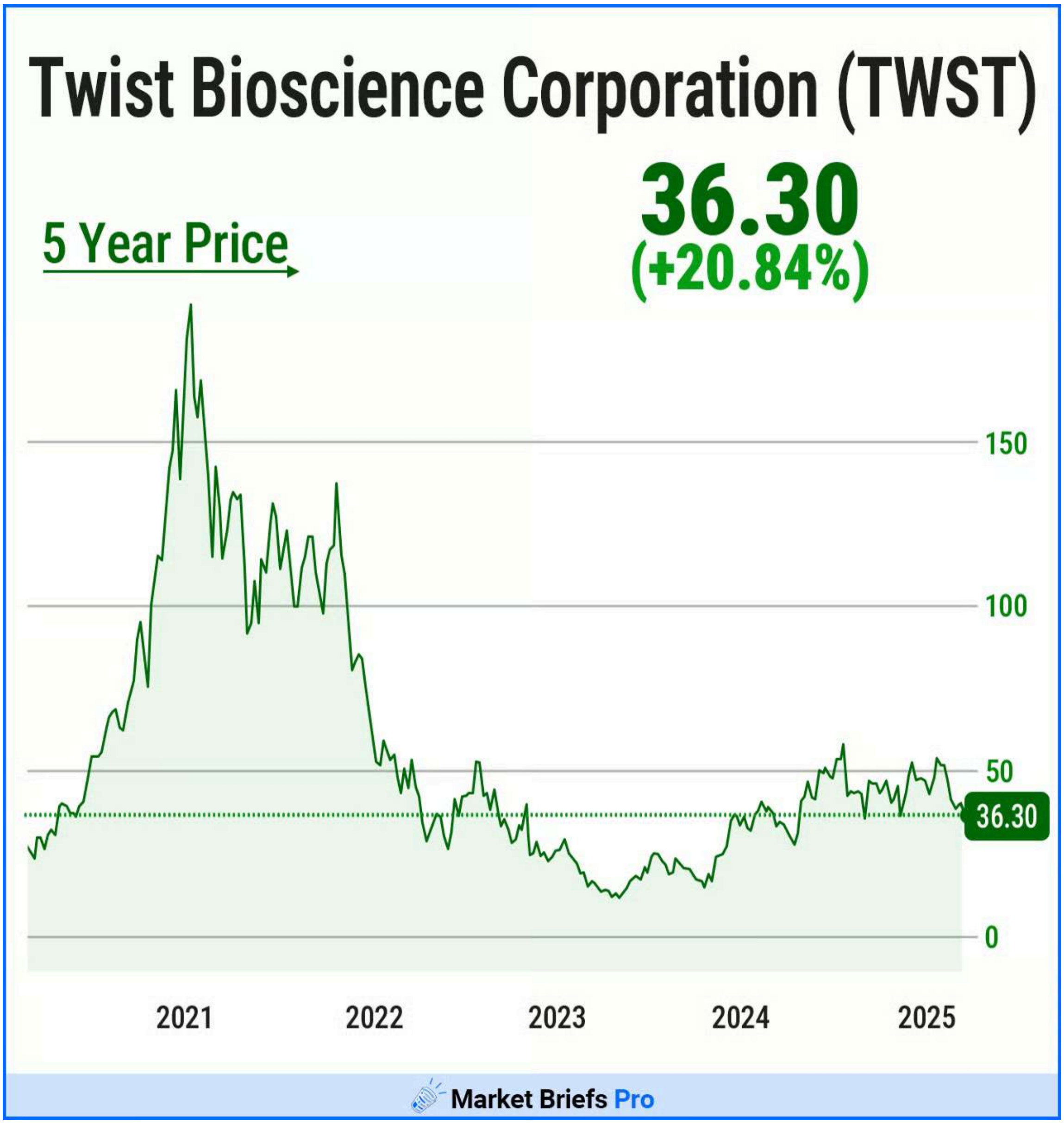
Fun fact: One person's DNA can store more data than all of the world's computers combined.

"It's not a stretch to think about a model that fully understands a DNA sequence and the chemistry behind it - the higher capacity of these GPUs now you can basically train a model on all the information in the world." - Aber Whitcomb.

Scientists are even trying to store data in DNA instead of using computers or data centers.

Twist Bioscience Corporation (TWST) is working with scientists to create this biomolecular code for molecular computing.

It focuses on man-made DNA and is helping companies like Microsoft build tiny tools that can find and kill diseases.

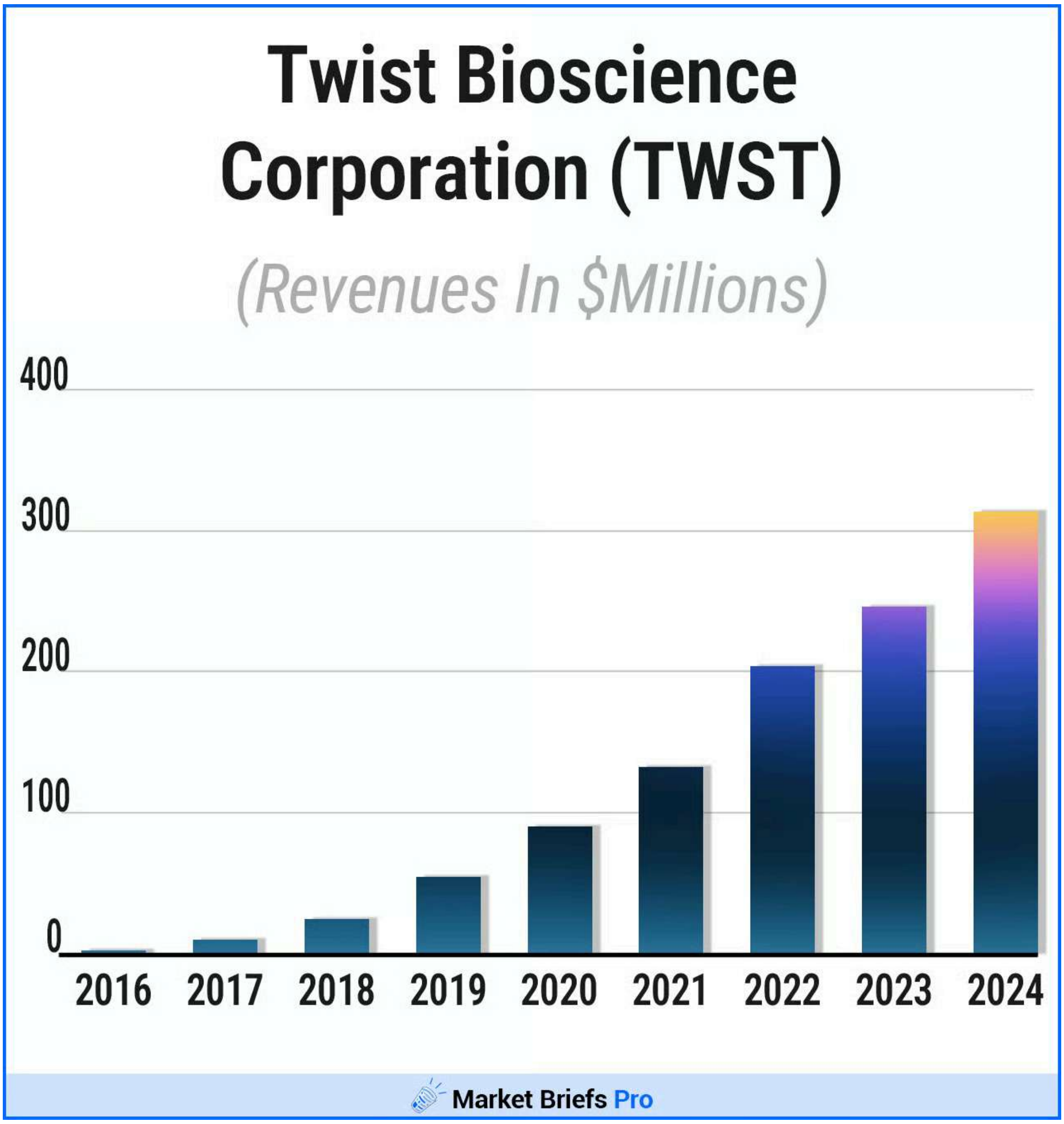


Data via Yahoo! Finance

In English? Twist uses chemicals that attach to our DNA via molecular computing.

By advancing this tech, Twist plays a critical role in molecular computing, creating libraries of synthetic DNA that could one day be used to cure diseases like cancer.

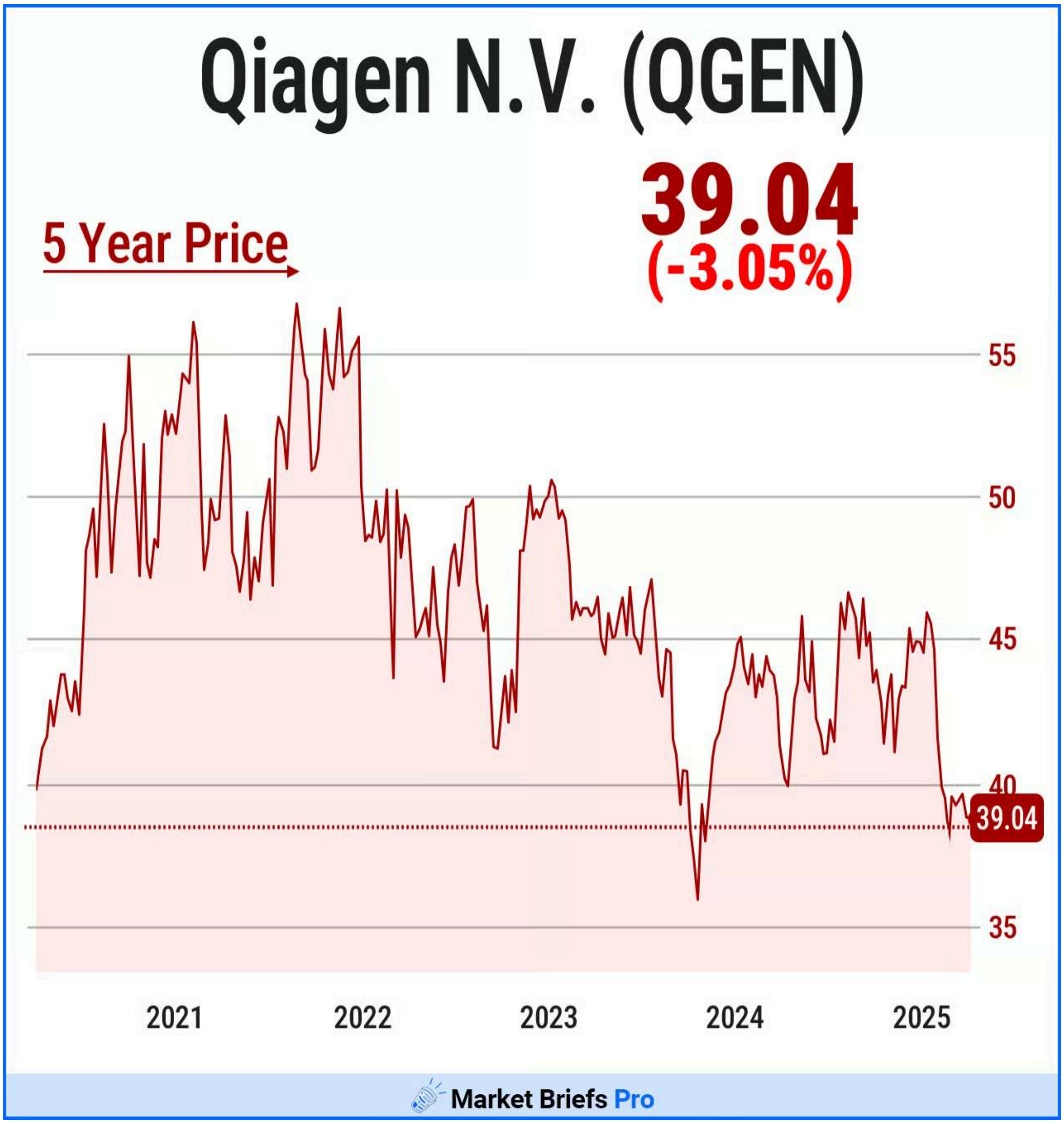
- Its revenue has more than doubled since 2020 as a result - jumping from $90 million in 2020 to over $300 million in 2024.



Data via company reports

On the other hand, the World Health Organization says 3 to 4 new diseases are found every year.

QIAGEN N.V. (QGEN) works to reverse engineer DNA and diseases at the molecular level by decoding the info found in the cells.

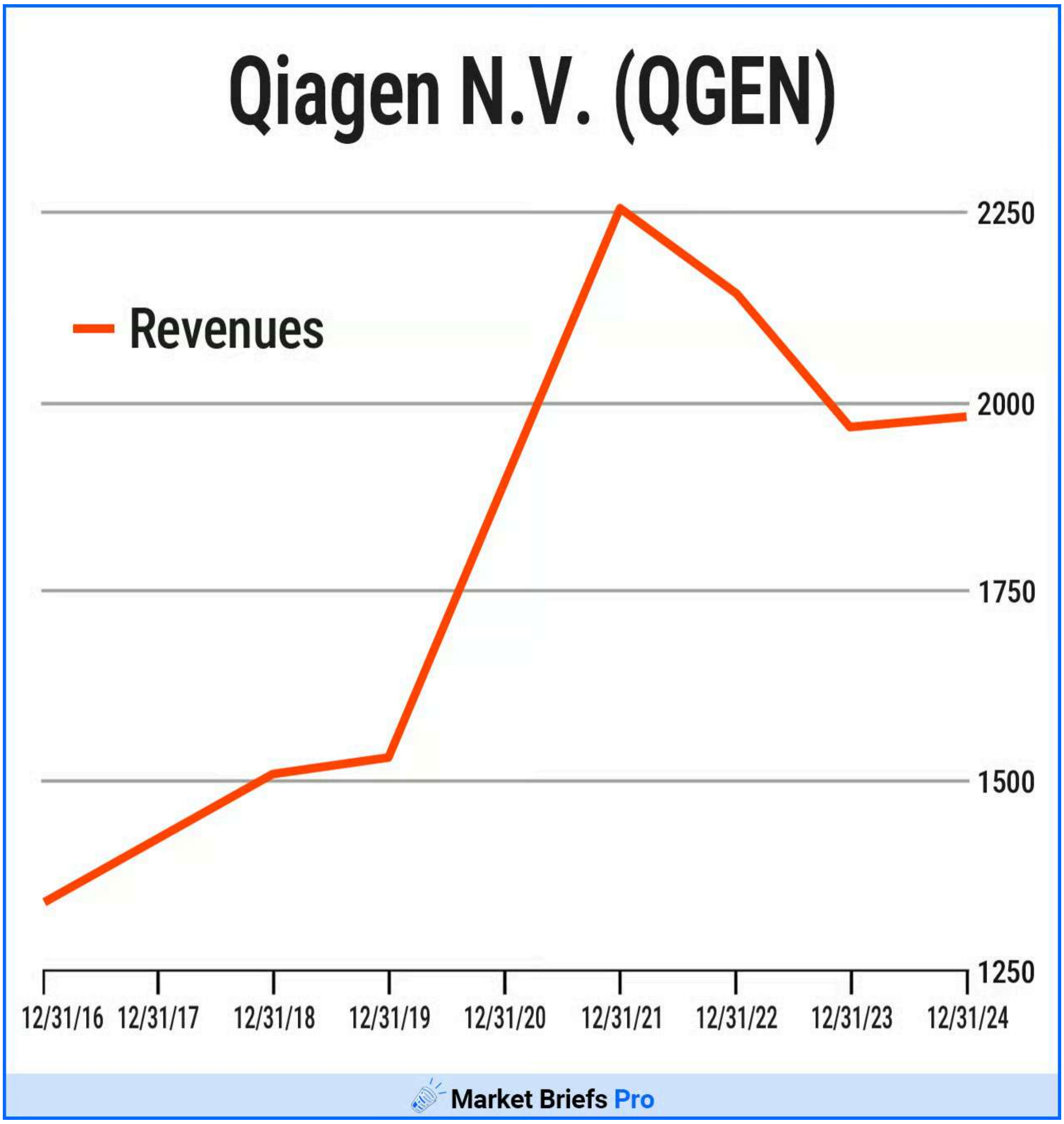


Data via Yahoo! Finance

It helps scientists find DNA no one has seen before - which could mean a new disease, virus, or DNA change.

Then, it breaks down exactly what it is so that researchers can work on cures.

QIAGEN is one of the global leaders in this space - it has over 500,000 customers worldwide and has uncovered 640 million discoveries in DNA sciences.

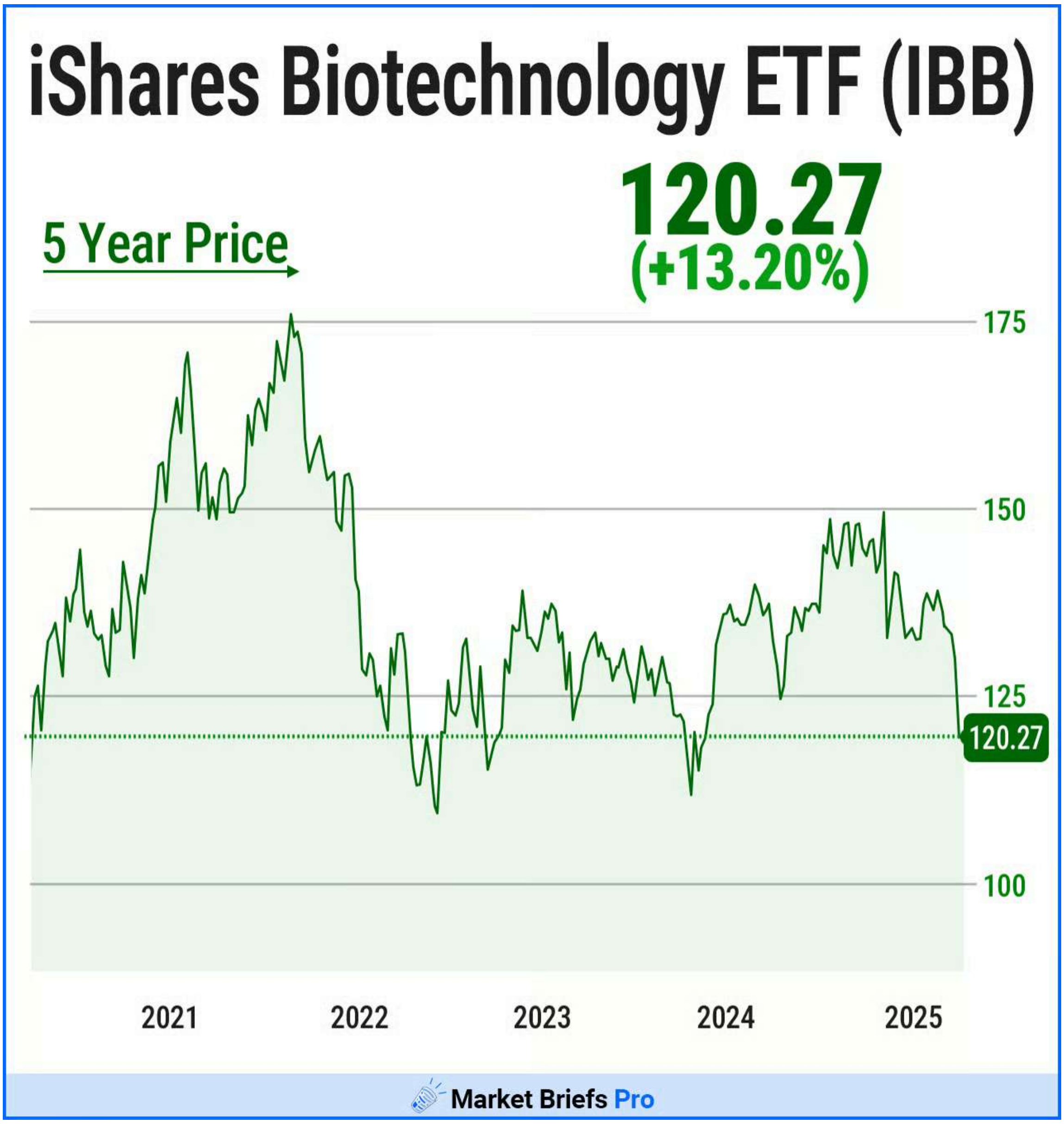


Data via company reports

- It also sells its tech as a kit that researchers can use on the go - as of March 2025, it has sold over 1 billion of these kits.

As molecular computing grows, QIAGEN is in a strong position to benefit from this shift

In 2024, it reported annual revenue of $1.97 billion, up from $1.96 billion in 2023.



Data via Yahoo! Finance

Opportunities in the molecular biology sector of molecular computing are expanding, including in the broader biotech market with ETFs like the iShares Biotechnology ETF.

CHEMICALS

Chemical Reactions

To make molecular computers work, companies will need special tools and chemicals.

Why? Remember your 7th grade science experiment? When you dropped Mentos into a bottle of Diet Coke?

That was a chemical reaction - molecular computers use these reactions to change and rewrite DNA.

In addition, it takes specialized tools and equipment to create molecular computers.

Why? Researchers often work with billions of molecules at once. The more molecules that are manipulated, the higher the probability of error.

"You need to kind of sit above all of these models and be able to take advantage of all the innovation happening there. From the lens of an investor the actual infrastructure layer on top of the model is where I would invest." - Aber Whitcomb

Agilent Technologies (A) is working to reduce errors by creating super powerful magnifying glasses.

These magnifying glasses can see something as tiny as a strand of DNA, making them more powerful than a microscope.

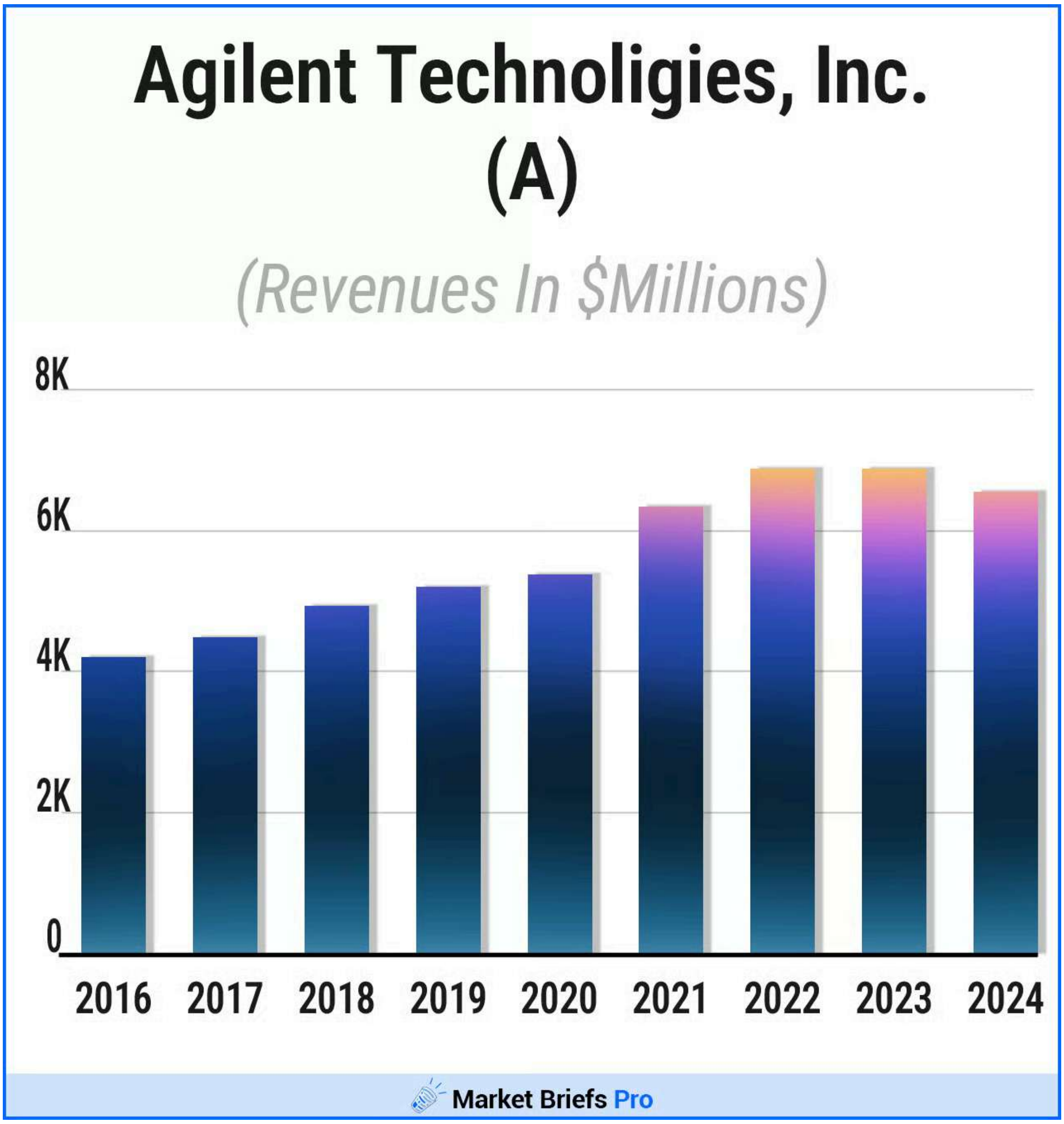
Data via Yahoo! Finance

- Its tech is so advanced that it can look within a piece of DNA and analyze it on the spot.

Researchers can then study diseases and cells at a molecular level.

- Agilent is on contract with tech giants like Nvidia and Amazon.

This tech has helped companies and scientists make their data move five times faster.

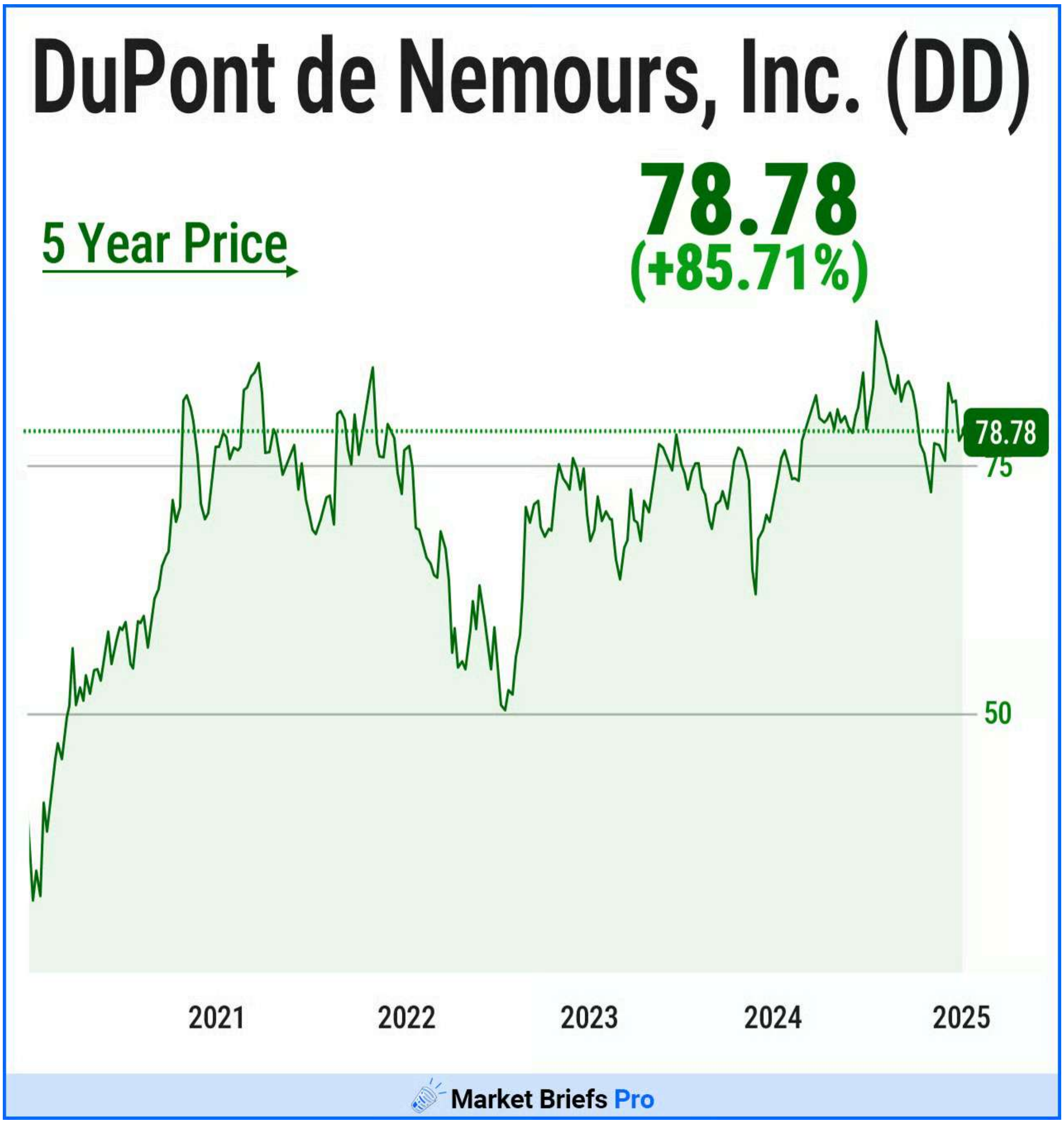


Data via Yahoo! Finance

- Agilent's Life Sciences and Applied Markets segment brought in $3.22 billion in revenue last year, alongside total revenue of $6.5 billion, which was near an all-time high.

On the chemicals side, DuPont de Nemours (DD) specifically is one of the world's largest chemical providers to labs.

DuPont creates films, chemicals, and materials that polish and flatten molecular computers.

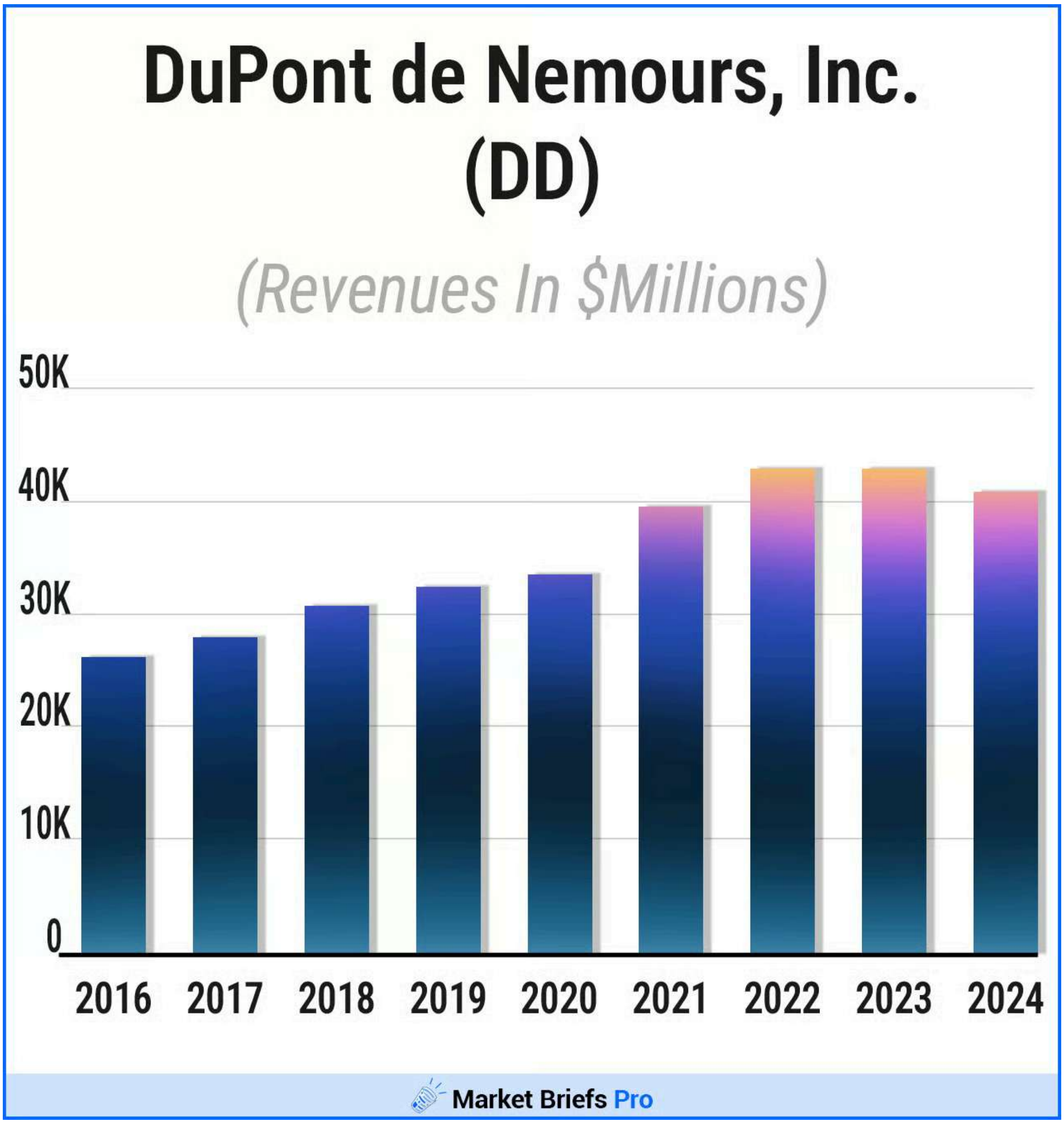


Data via Yahoo! Finance

Why is this important? Many of these molecular computers are designed to go inside humans, which means they need to be safe.

Jagged, sharp, or dirty objects, even tiny ones, can cause humans harm.

This means that molecular computers must be coated in special chemicals before use, something that DuPont specializes in.



Data via Yahoo! Finance

Financially, DuPont merged with Dow chemicals back in 2017, before splitting again in 2019.

As the company restructures, its revenue has remained flat - around $12 billion for the last 5 years.

Still, by providing the vast majority of molecular computing labs around the world with the chemicals they need to operate, DuPont is well positioned to benefit from this shift in the long term.

RISKS

Small Issues

While molecular computing is creating some long-term opportunities, investors should be aware of a few key things:

First, it may take a decade or more for these companies to realize the gains from their innovations.

- Molecular computers still need a lot of research - spending is growing, but it’s still a small market right now.

Second, even though it’s cheaper to make a molecular computer now, it’s still pretty expensive.

- Quantum Insider estimates that the average molecular computer costs around $10 million to build.

Lastly, it might take years before molecular computers start making money. For now, companies might lose a lot of money because researching this kind of tech takes a lot of time and cash.

This makes it risky for investors because no one knows if people will actually use molecular computers in the future.

"So some of the use cases that have been (I wouldn't say proven out but we've got solid proof of concepts) are more in storage, and I think one of the attractions of molecular computing is you can just sort of store a lot more data in a smaller space with lower energy consumption in DNA than you might somewhere else." - Nate Beyor

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REVIEW

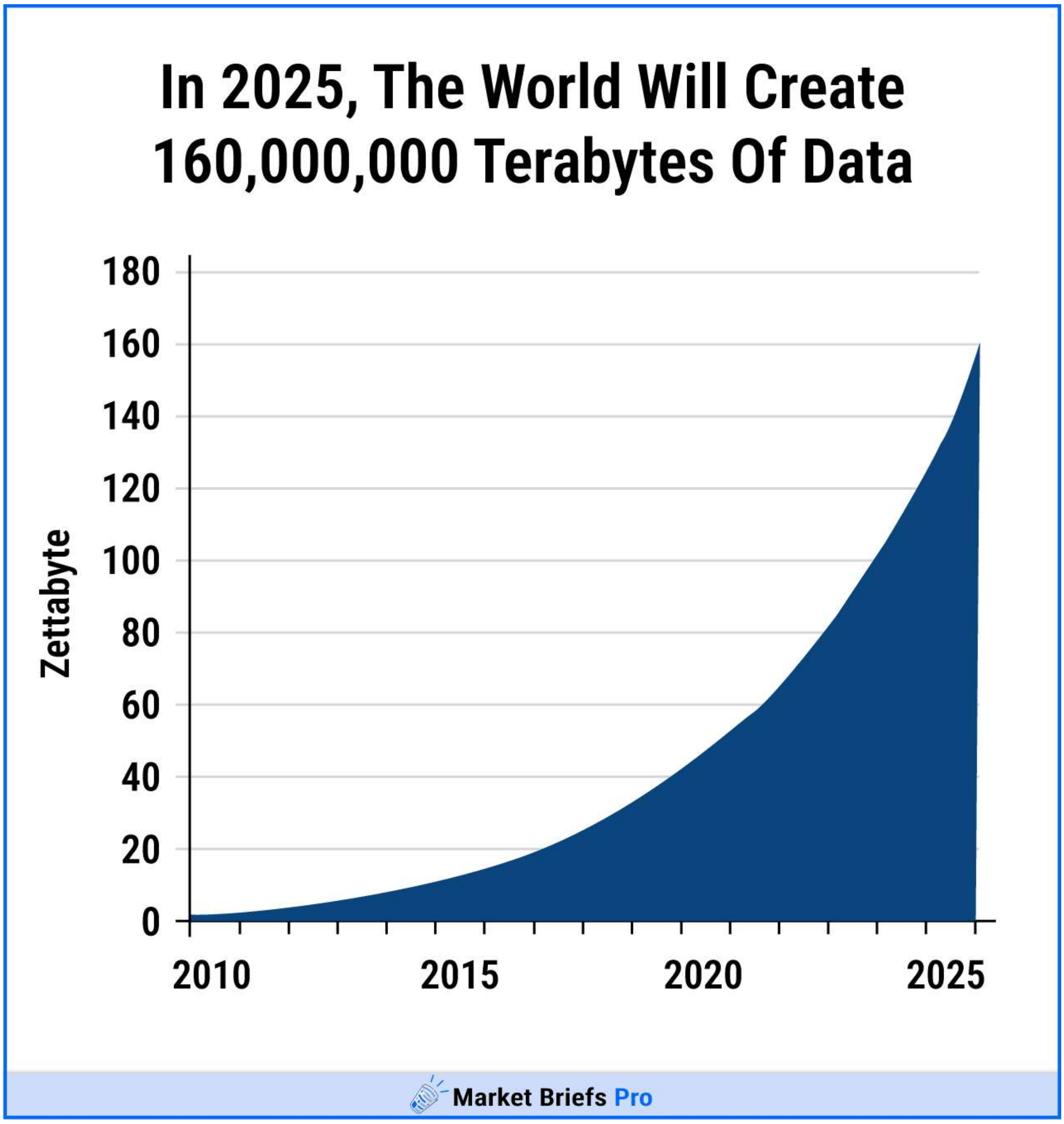
Tiny Innovations, Big Opportunities

Molecular computers are easier to use now than ever before.

That’s changing how scientists and businesses think about medicine and data.

Because of this, the market is now expanding. Nanotech, molecular biology, chemical, and equipment firms in this space will benefit from this shift.

And remember - humans have more data than all of the world’s computers combined.



Data via Big Data Wire

But this change is still very new, so investors should be ready to wait a long time.

"I think we're still in an era of picks and shovels to a certain extent.
It's important for investors to be realistic around, what success will look like for these companies and, how quickly they're going to realize it." - Nate Beyor

No one knows for sure if molecular computers will become big, but their tiny ideas are already making a splash in tech.

Investors will want to watch out for developments from this industry over the next few years.

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